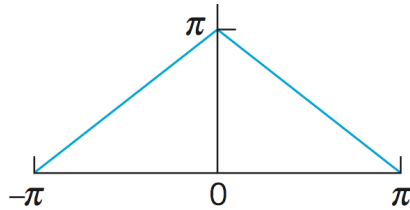


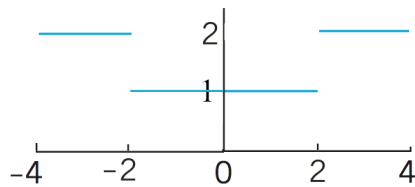
Answer the following and show the integration details! Don't just plug it into a CAS.

**Problem 1:** Determine the Fourier Series for each of the periodic functions depicted in the graphs below. Assume each of them periodically extend along the  $x$ -axis.

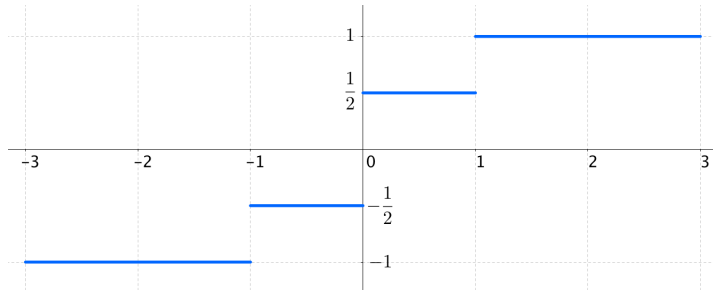
(i)



(ii)



(iii)



**Problem 2:** Let  $f(x) = x^2$  on  $-1 \leq x \leq 1$  and define  $f(x)$  to have period 2 ( $f(x) = f(x+2)$ ). Show that the Fourier Series for  $f(x)$  is given by:

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi x)$$

**Extra Credit:** Use the result of **problem 2** to derive the summation identities

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$